ECEN – 260 Final Project Design

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BYU-IDAHO | Microprocessor-based System Designs

This project is a hydroponics/greenhouse environmental monitoring system designed to meet the need for accurate and timely monitoring of growing conditions. In hydroponic and greenhouse systems, plant health depends on maintaining proper conditions such as temperature, humidity, and water level. If these conditions are not monitored carefully, crops can be damaged, growth can become inconsistent, and resources can be wasted. This system collects data from sensors, processes that data, and displays it in a clear and understandable format so the user can make better decisions about maintaining the growing environment. In the future, the system could be expanded to include devices such as fans and water pumps, along with additional sensors such as pH and particulate sensors.

This design also considers public health, safety, welfare, and broader global, cultural, social, environmental, and economic factors. From a health and welfare perspective, improved monitoring supports healthier crop production by maintaining stable conditions and reducing the risk of poor-quality yields. From a safety perspective, monitoring water levels and environmental conditions can help prevent failures that could damage equipment or create hazards. Environmentally, the system promotes efficient use of water and other resources, which is one of the major benefits of hydroponics. Economically, it can reduce crop loss and improve productivity by helping users identify problems early. On a broader level, systems like this can support more sustainable agriculture and more reliable food production in different communities and settings.

Three major course concepts integrated into this design are analog-to-digital conversion (ADC), LCD display interfacing, and external interrupts. ADC allows analog sensor signals to be converted into digital data that the microcontroller can process. LCD interfacing makes it possible to present sensor readings in a clear and readable way for the user. External interrupts improve system responsiveness by allowing important events to be handled immediately instead of relying only on constant polling. In addition, the DHT11 sensor demonstrates digital communication by sending temperature and humidity data directly to the controller. There is also an RGB LED using PWM to be an additional indicator to the user of the temperature level of the system. Together, these concepts create a practical and effective environmental monitoring system.